LESSON TWO

GITHUB CONCEPTS AND TEAM COLLABORATION

2.1 Important GitHub Concepts

I learned that GitHub has different features that help developers manage projects and work together.

1. Commit

A commit is a saved record of changes made to a project. Whenever I make important changes to my files, I can create a commit to save those changes in the project's history.

Example:

```
git commit -m "Added login page"
```

The message explains the changes I have saved.

2. Fork

I learned that forking means creating a personal copy of another person's repository under my GitHub account. It allows me to work on the copied project without directly changing the original repository.

For example, if another developer has a project on GitHub and I want to contribute to it, I can fork the repository and work on my own copy.

3. Pull Request

A pull request is a request to merge changes from one branch or repository into another. It allows other developers to review my contributions before they are accepted into the main project.

4. Branch

A branch is a separate line of development within a repository. It allows me to work on a new feature without directly changing the main version of the project.

For example:

```
git branch login-feature
```

This creates a new branch called login-feature.

5. Push

Push means uploading my local commits from my computer to a remote repository on GitHub.

6. Pull

Pull means downloading changes from a remote repository and integrating them into my local working branch.

7. Clone

Clone means creating a local copy of an existing remote repository on my computer.

2.2 Team Collaboration on GitHub

I learned that GitHub makes it possible for a group of developers to work together on one software project. Each member can contribute to the project based on their assigned responsibilities.

For example, in a website development project:

- One member can develop the homepage.
- Another member can develop the login page.
- Another member can work on the dashboard.
- Another member can handle the database-related files.

After completing their tasks, members can upload their contributions to GitHub.

2.3 Responsibilities of a Team Lead

I learned that a team lead is the person responsible for coordinating the activities of the team. The team lead ensures that every member understands their responsibilities and that the project is progressing properly.

Some responsibilities of a team lead include:

1. Creating the main project repository.
2. Inviting team members to collaborate.
3. Assigning tasks to members.
4. Monitoring the progress of the project.
5. Reviewing pull requests.
6. Helping members solve technical problems.
7. Ensuring that the project is properly organized.

2.4 Responsibilities of Team Members

I learned that team members are expected to complete their assigned tasks and contribute positively to the project.

Their responsibilities include:

1. Developing their assigned features.
2. Creating commits for their changes.
3. Pushing their work to GitHub.
4. Following the team's coding standards.
5. Communicating with other members.
6. Correcting errors identified during code review.

2.5 How Team Collaboration Works

A typical team collaboration process involves the following steps:

1. The team lead creates a repository on GitHub.
2. The team lead invites other developers as collaborators.
3. Each member receives an assigned task.
4. Members create branches for their tasks.
5. Members work on their assigned files.
6. They commit and push their changes.
7. They create pull requests.
8. The team lead or another authorized member reviews the changes.
9. Approved changes are merged into the main branch.

Conclusion

From this lesson, I understood that GitHub provides important features such as commits, forks, pull requests and branches. I also learned how developers can use GitHub to collaborate, share responsibilities and complete software projects as a team.